

```
In [8]: import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
```

```
In [9]: df
```

Out[9]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size
0	1	55	Male	Blouse	Clothing	53	Kentucky	
1	2	19	Male	Sweater	Clothing	64	Maine	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	
4	5	45	Male	Blouse	Clothing	49	Oregon	
...	...	...	...	...	...	...	...	...
3895	3896	40	Female	Hoodie	Clothing	28	Virginia	
3896	3897	52	Female	Backpack	Accessories	49	Iowa	
3897	3898	46	Female	Belt	Accessories	33	New Jersey	
3898	3899	44	Female	Shoes	Footwear	77	Minnesota	
3899	3900	52	Female	Handbag	Accessories	81	California	

3900 rows × 18 columns

```
In [5]: df.head(10)
```

Out[5]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size
0	1	55	Male	Blouse	Clothing	53	Kentucky	L
1	2	19	Male	Sweater	Clothing	64	Maine	L
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M
4	5	45	Male	Blouse	Clothing	49	Oregon	M
5	6	46	Male	Sneakers	Footwear	20	Wyoming	M
6	7	63	Male	Shirt	Clothing	85	Montana	M
7	8	27	Male	Shorts	Clothing	34	Louisiana	L
8	9	26	Male	Coat	Outerwear	97	West Virginia	L
9	10	57	Male	Handbag	Accessories	31	Missouri	M

In [6]: `df.info()`

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   str
3   Item Purchased                      3900 non-null   str
4   Category                            3900 non-null   str
5   Purchase Amount (USD)               3900 non-null   int64
6   Location                            3900 non-null   str
7   Size                                3900 non-null   str
8   Color                               3900 non-null   str
9   Season                              3900 non-null   str
10  Review Rating                       3863 non-null   float64
11  Subscription Status                 3900 non-null   str
12  Shipping Type                      3900 non-null   str
13  Discount Applied                   3900 non-null   str
14  Promo Code Used                    3900 non-null   str
15  Previous Purchases                  3900 non-null   int64
16  Payment Method                     3900 non-null   str
17  Frequency of Purchases              3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 835.9 KB
```

In [9]: `df.describe()`

Out[9]:

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

In [10]: df.tail()

Out[10]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size
3895	3896	40	Female	Hoodie	Clothing	28	Virginia	L
3896	3897	52	Female	Backpack	Accessories	49	Iowa	L
3897	3898	46	Female	Belt	Accessories	33	New Jersey	L
3898	3899	44	Female	Shoes	Footwear	77	Minnesota	S
3899	3900	52	Female	Handbag	Accessories	81	California	M

In [12]: df.shape

Out[12]: (3900, 18)

In [15]: df.columns

Out[15]: Index(['Customer ID', 'Age', 'Gender', 'Item Purchased', 'Category', 'Purchase Amount (USD)', 'Location', 'Size', 'Color', 'Season', 'Review Rating', 'Subscription Status', 'Shipping Type', 'Discount Applied', 'Promo Code Used', 'Previous Purchases', 'Payment Method', 'Frequency of Purchases'], dtype='str')

In [18]: df.describe(include='all')

Out[18]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	
unique	NaN	NaN	2	25	4	NaN	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Mor
freq	NaN	NaN	2652	171	1737	NaN	
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	

In [20]: `df.isnull().sum()`

Out[20]:

Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	37
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Payment Method	0
Frequency of Purchases	0

dtype: int64

In [24]: `df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x - x.mean())`

In [25]: `df.isnull().sum()`

```
Out[25]: Customer ID      0
        Age              0
        Gender           0
        Item Purchased   0
        Category         0
        Purchase Amount (USD) 0
        Location         0
        Size             0
        Color            0
        Season           0
        Review Rating     0
        Subscription Status 0
        Shipping Type     0
        Discount Applied  0
        Promo Code Used   0
        Previous Purchases 0
        Payment Method    0
        Frequency of Purchases 0
        dtype: int64
```

```
In [33]: df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)' : 'purchase_amount'})
df.columns
```

```
Out[33]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
               'purchase_amount', 'location', 'size', 'color', 'season',
               'review_rating', 'subscription_status', 'shipping_type',
               'discount_applied', 'promo_code_used', 'previous_purchases',
               'payment_method', 'frequency_of_purchases'],
              dtype='str')
```

```
In [35]: # create a column age_group
labels = ['Young adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)
```

```
In [37]: df[['age', 'age_group']].head(5)
```

```
Out[37]:
```

	age	age_group
0	55	Middle-aged
1	19	Young adult
2	50	Middle-aged
3	21	Young adult
4	45	Middle-aged

```
In [39]: # create a column purchase_frequency_days

frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
}
```

```

    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_map)

```

```
In [41]: df[['purchase_frequency_days', 'frequency_of_purchases']].head(10)
```

```
Out[41]:
```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly
8	365	Annually
9	90	Quarterly

```
In [43]: df[['discount_applied', 'promo_code_used']].head(10)
```

```
Out[43]:
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
In [45]: (df['discount_applied'] == df['promo_code_used']).all()
```

```
Out[45]: np.True_
```

```
In [48]: df = df.drop('promo_code_used', axis = 1)
```

```
In [49]: df.columns
```

```
Out[49]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
              'purchase_amount', 'location', 'size', 'color', 'season',
              'review_rating', 'subscription_status', 'shipping_type',
              'discount_applied', 'previous_purchases', 'payment_method',
              'frequency_of_purchases', 'age_group', 'purchase_frequency_day
              s'],
              dtype='str')
```

```
In [15]: import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
```

```
In [16]: df
```

```
Out[16]:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size
0	1	55	Male	Blouse	Clothing	53	Kentucky	
1	2	19	Male	Sweater	Clothing	64	Maine	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	
4	5	45	Male	Blouse	Clothing	49	Oregon	
...	...	...	...	...	...	...	...	...
3895	3896	40	Female	Hoodie	Clothing	28	Virginia	
3896	3897	52	Female	Backpack	Accessories	49	Iowa	
3897	3898	46	Female	Belt	Accessories	33	New Jersey	
3898	3899	44	Female	Shoes	Footwear	77	Minnesota	
3899	3900	52	Female	Handbag	Accessories	81	California	

3900 rows × 8 columns

```
In [23]: pip install --user psycopg2-binary sqlalchemy
```

```
Requirement already satisfied: psycopg2-binary in .\AppData\Roaming\Python
\Python314\site-packages (2.9.13)
Requirement already satisfied: sqlalchemy in C:\ProgramData\anaconda3\Lib
\site-packages (2.0.51)
Requirement already satisfied: greenlet>=1 in C:\ProgramData\anaconda3\Lib
\site-packages (from sqlalchemy) (3.3.2)
Requirement already satisfied: typing-extensions>=4.6.0 in C:\ProgramData
\anaconda3\Lib\site-packages (from sqlalchemy) (4.16.0)
Note: you may need to restart the kernel to use updated packages.
```

```

In [28]: import pandas as pd
from sqlalchemy import create_engine
from sqlalchemy.engine import URL

# Step 1: Define connection parameters
connection_url = URL.create(
    drivename="postgresql+psycopg2",
    username="postgres",
    password="Fareed@123", # Handles the '@' automatically
    host="localhost",
    port=5432,
    database="customer_behavior",
)

# Create connection engine
engine = create_engine(connection_url)

# Step 2: Load DataFrame into PostgreSQL
table_name = "customer"

with engine.connect() as connection:
    df.to_sql(
        name=table_name,
        con=connection,
        if_exists="replace",
        index=False,
        chunksize=1000,
    )

print(
    f"Data successfully loaded into table '{table_name}' in database"
    f" '{connection_url.database}'."
)

```

Data successfully loaded into table 'customer' in database 'customer_behavior'.

In []: